

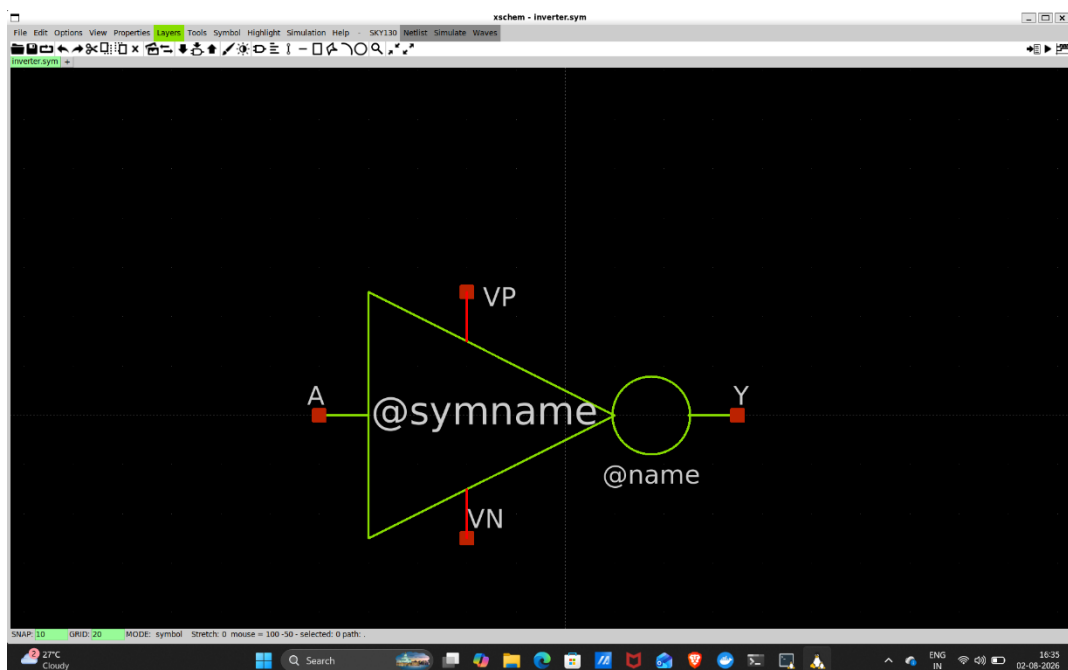
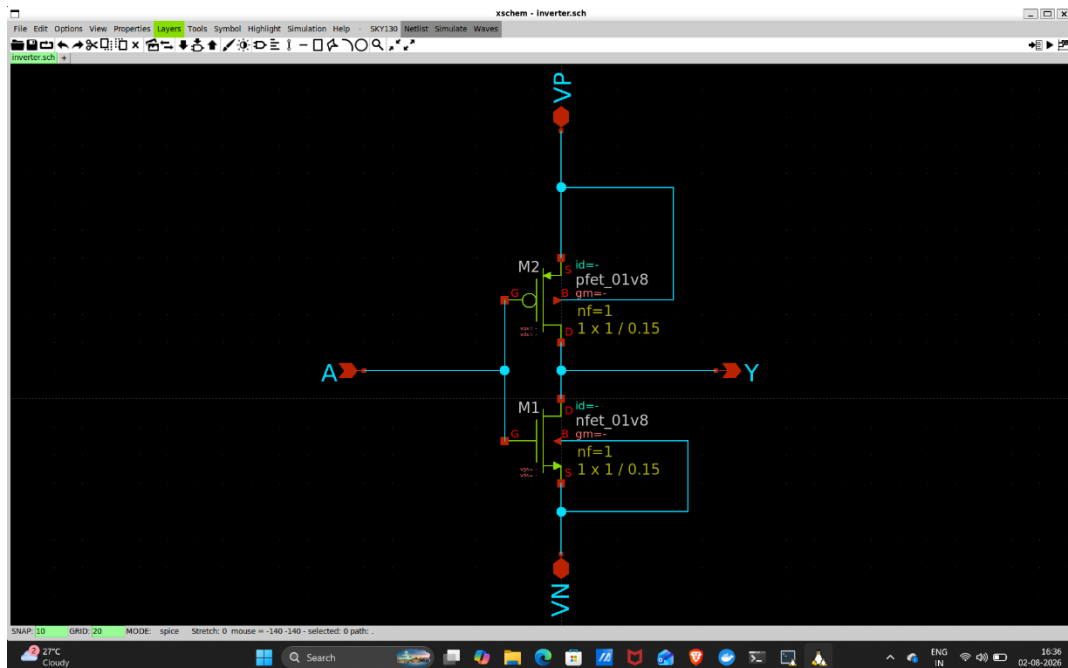
CMOS Inverter – Using sky130A PDK

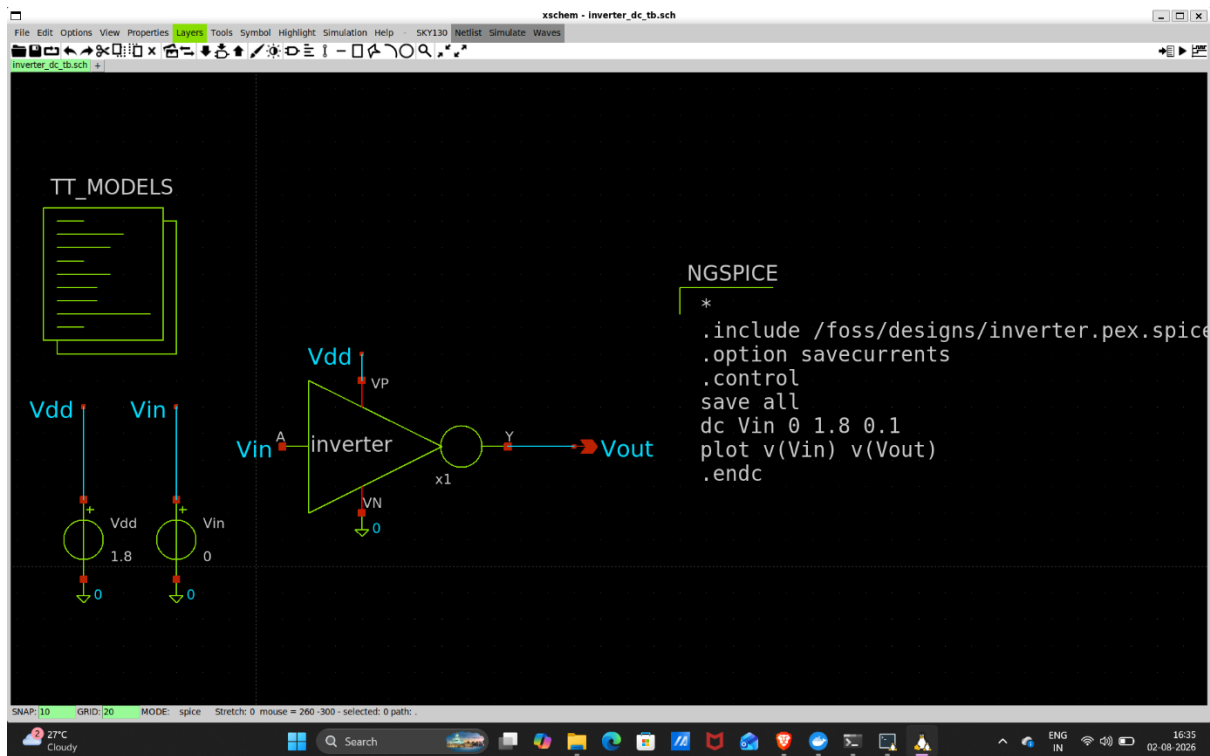
[via IIC-OSIC-TOOLS]

-ARJUN RAJ

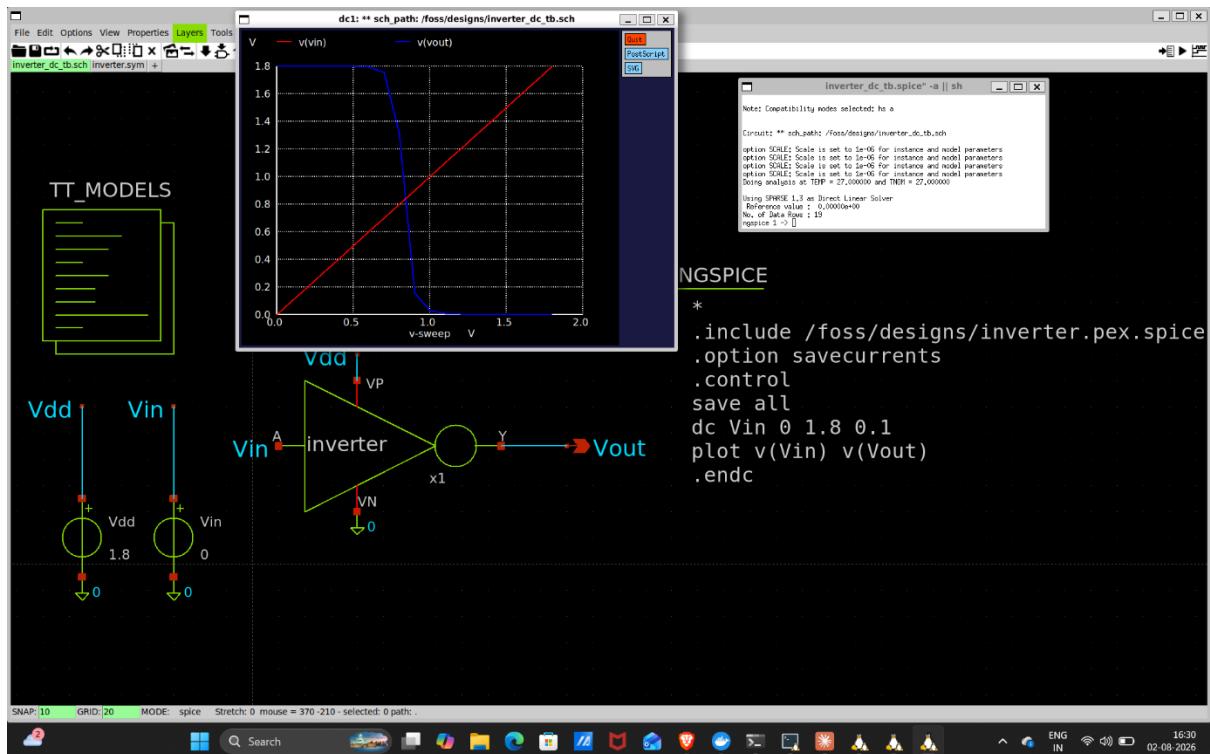
2025bec0025

Xschem





Ngspice Simulations





Verification

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File Edit View Terminal Tabs Help
Cell pin lists are equivalent.
Device classes inverter and inverter are equivalent.

Final result: Verify: cell /foss/designs/inverter_magic.spice has no elements and/or nodes. Not checked.
/foss/designs > magic -t sky130a.tech inverter.mag
/foss/designs > echo SPDK
ihp-sgl3g2
/foss/designs > export PDK=sky130a
/foss/designs > echo SPDK
sky130a
/foss/designs > iic-lvs.sh -d -s inverter.sch -l inverter.mag -c inverter
[INFO] DEBUG is enabled!
[INFO] flag -s is set to <inverter.sch>.
[INFO] flag -l is set to <inverter.mag>.
[INFO] flag -c is set to <inverter>.
[INFO] sky130 PDK selected.
[INFO] Using schematic file </foss/designs/inverter.sch>.
[INFO] Using layout file </foss/designs/inverter.mag>.
[INFO] Running LVS of </foss/designs/inverter.mag> vs </foss/designs/inverter.sch>.
[INFO] Extracting SPICE netlist from schematic </foss/designs/inverter.sch>...
[INFO] Extracting netlist from layout </foss/designs/inverter.mag>...
[INFO] Run netgen...
...
[INFO] MagicNetgen LVS is OK, schematic/netlist and layout match!
...
CONGRATULATIONS! LVS is OK, schematic/netlist and layout match!

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  /  \
 /____\

[DONE] Bye!
/foss/designs > iic-px.sh

PEX script using Magic (ICD@JKU)

Usage: /foss/tools/sak/iic-px.sh [-d] [-m <mode>] [-s <mode>] [-n <subcname>] [-w <workdir>]
      [-t <threshold>] [-r <minres>] [-y <mindelay>] <cellname>

  -m Select PEX mode (1 = C-decoupled, 2 = C-coupled [default], 3 = full-RC)
  -s Subcircuit definition in PEX netlist (1 = include subcircuit definition [default], 0 = no subcircuit)
  -n Name of PEX subcircuit (default is <cellname>)
  -w Use <workdir> to store result files (default current dir)
  -t full-RC only: extresist threshold in mOhm (default 10000 = 10 Ohm)
  -r full-RC only: extresist minres in mOhm (default 1000 = 1 Ohm)
  -y full-RC only: extresist mindelay in ps (default 1, 0 = gate by resistance instead of delay)
  -d Enable debug information

<cellname> may be a cell name or a layout file (.mag, .mag.gz, .gds, .gds.gz)
NOTE: for a GDS input the top cell must be named like the file (<cellname>)

/foss/designs >
```

```
Brave Google
Microsoft Edge
IMA211...
VLC media player
Visual Studio Code
WhatsApp Web
Discord LMS
Recycle Bin Zoom Workplace

[arjun_raj@Union: ~/iic-asic-t] x + -
[HINT] It can also be restarted with "docker start iic-asic-tools_xserver_uid_1000" or removed with "docker rm iic-asic-tools_xserver_uid_1000" if required.

Press "s" to start, and "r" to remove: s
iic-asic-tools_xserver_uid_1000
socat is still running. Press Ctrl+C to stop it.
WARNING: This will kill a running container!
Container iic-asic-tools_xserver_uid_1000 is no longer running.
Stopping socat...
2026/08/02 09:48:01 socat[860] W exiting on signal 15
socat stopped.
socat or the container is no longer running. Exiting...
Stopping socat...
socat stopped.
arjun_raj@Union:~/iic-asic-tools$ bash start_x.sh
[INFO] Container engine auto-set to docker.
[INFO] Design directory auto-set to /home/arjun_raj/eda/designs
[INFO] Auto detected Linux.
Starting socat to enable TCP connections to the X-Server for indirect rendering.
Started socat with PID 162105 in the background..
[INFO] Using indirect rendering.
[WARNING] Container iic-asic-tools_xserver_uid_1000 exists
[HINT] It can also be restarted with "docker start iic-asic-tools_xserver_uid_1000" if required.

Press "s" to start, and "r" to remove: s
iic-asic-tools_xserver_uid_1000
socat is still running. Press Ctrl+C to stop it.
WARNING: This will kill a running container!

File Edit View Terminal Tabs Help
[INFO] DEBUG is enabled!
[INFO] flag -b is set.
[INFO] ihp-sgl3g2 PDK selected.
[INFO] Using layout file <inverter.mag>.
[WARNING] Layout DRC needs a GDS layout, running Magic DRC only.
[INFO] Running DRC of <inverter.mag>.
[INFO] Results are put into </foss/designs>.
[INFO] Launching Magic DRC...
[INFO] Magic runs DRC on the .mag file.
...
[INFO] Magic DRC is clean!
...
CONGRATULATIONS! No DRC errors in <inverter> found!

  (0) (0)
   <
  /  \
 /____\

[DONE] Bye!
/foss/designs >
```